

Framework for Assessing Changes To Sea-level **(FACTS)**

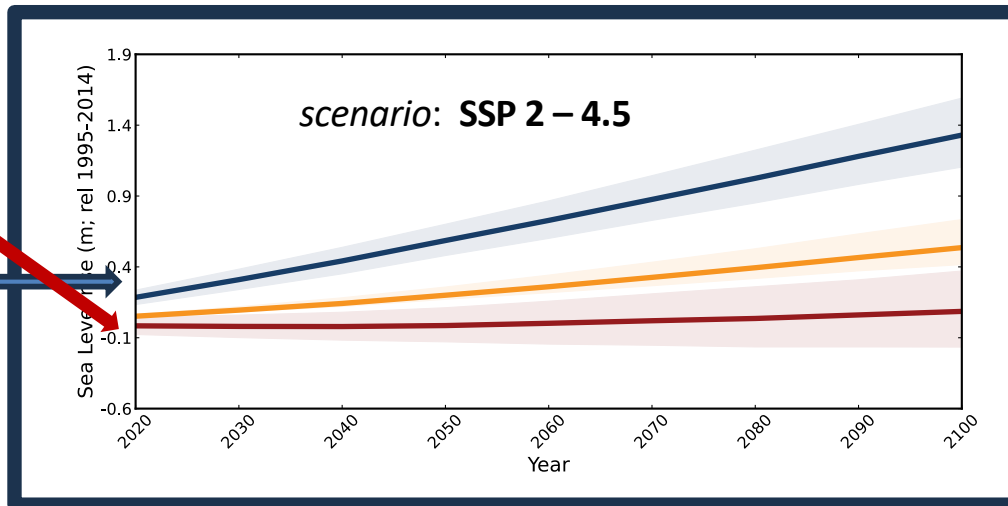
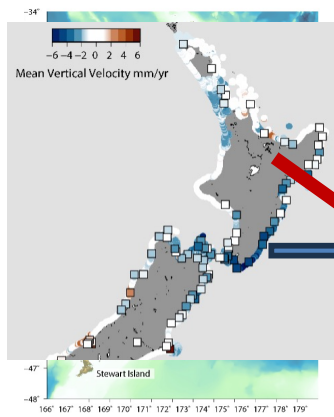
A Modular Probabilistic Sea-Level Projection Framework

Praveen Kumar

Contents:

Introduction : Probabilistic Sea-level Projections Using Framework for Assessing Changes To Sea-level (FACTS)

- ✓ **Section 1. Deep Uncertainty, Ambiguity & FACTS**
Scenario who?? SSP what?
Icesheet and Emission Deep Uncertainty
- ✓ **Section 2. Uncertainty Communication IPCC assessment reports**
- ✓ **Section 3. Tool to Assess & Communicate Uncertainty Framework
for Assessing Changes To Sea-level (FACTS)**



Climate Step	fair/temperature		
Sea Level Components Step	emulandice/AIS	emulandice/GrlS	emulandice/glaciers
	larmip/AIS	FittedISMIP/GrlS	ipccar5/glaciers
	ipccar5/AIS	deconto21/AIS	bamber19/icesheets
	tlm/sterodynamics	kopp14/verticallandmotion	ssp/landwaterstorage
Integration Step	facts/total		
	WF1e	WF1f	WF2e
	WF2f	WF3e	WF3f
	WF4		



Framework for **A**ssessing **C**hanges **T**o **S**ea-level
IPCC AR6 style Probabilistic Projections.

FACTS

Goal: *Constrained Characterization of Sea-level Uncertainty to support climate risk management.*



RUTGERS

School of Arts and Sciences
DEPARTMENT OF EARTH AND
PLANETARY SCIENCES

Earth System Science & Policy Lab



Prof. Bob Kopp

We addresses four interrelated questions:



How has sea level changed in the past?



How may sea level change in the future in response to climate forcing?



How do climate and sea-level change impact the economy and human well-being?



How can climate and sea-level science more effectively support climate risk management under deep uncertainty?



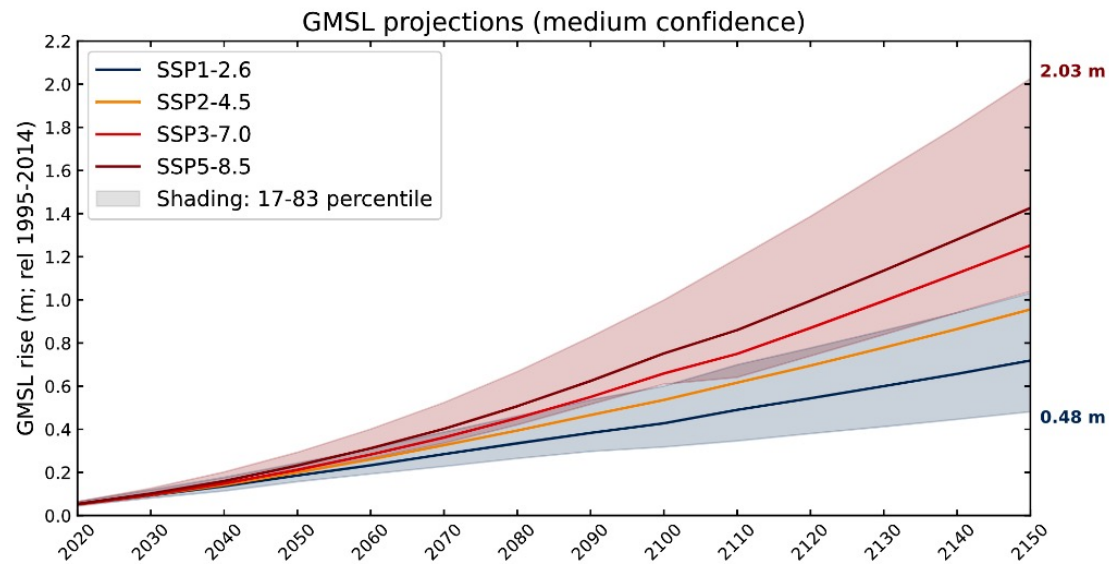
SEA LEVEL CHANGE
Observations from Space



Introduction :

**Probabilistic Sea-level Projections
Using FACTS**

An Example of Probabilistic Sea-level Projection



Also See: <https://www.ipcc.ch/report/ar6/wg1/figures/chapter-9/figure-9-27/>

Researchers spend **A lot** of time, effort & resources to understand and characterize the various sources of uncertainty **with a goal of reducing it.**

Why bother reducing uncertainty in sea-level projections ?

*Smaller **uncertainty means** better risk assessment and proactive adaptation reduce long-term damage and emergency repair costs.*

Forecast *what we expect*

A look into the future based on current trends, historical data, and realistic expectations the most likely outcome on the current trajectory.

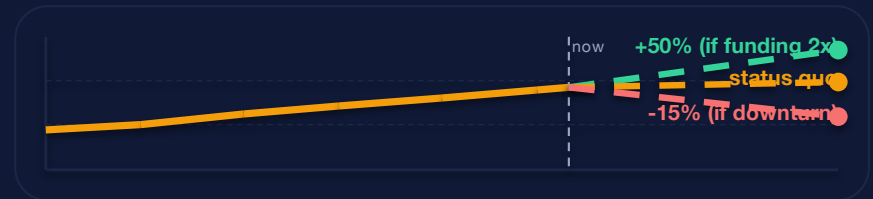


A **forecast** is a single best guess.

Projection

“What could happen if X occurs”

A look into the future based on hypothetical scenarios and assumptions — what is possible under specific conditions, not necessarily what is most likely.



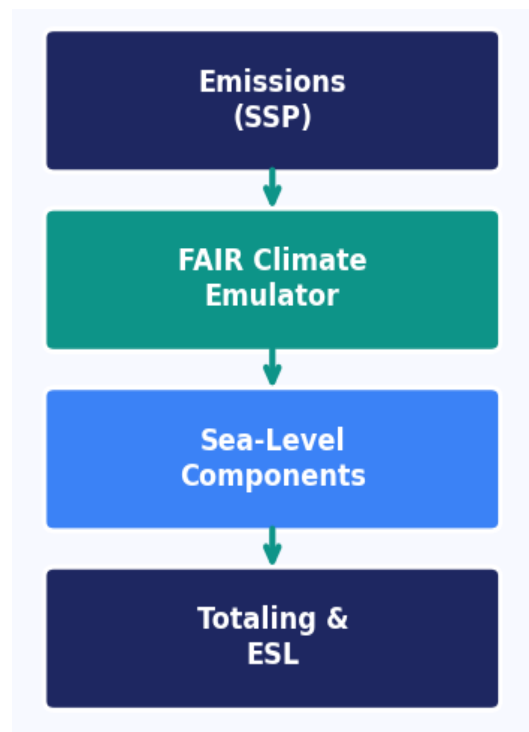
A **projection** is a bunch of possibilities.

What is FACTS

Framework for Assessing Changes To Sea-level?

- Open-source, modular **framework** for probabilistic sea-level change projections
- IPCC AR6 assessment; produces global and location-specific sea-level estimates (2020 to 2150/2300)
- Final output enables risk assessments, coastal planning, and policy decisions

The Framework for Assessing Changes To Sea-level (FACTS) v1.0: a platform for characterizing parametric and structural uncertainty in future global, relative, and extreme sea-level change



Robert E. Kopp ✉, Gregory G. Garner, Tim H. J. Hermans, Shantenu Jha, Praveen Kumar, Alexander Reedy, Aimée B. A. Slangen, Matteo Turilli, Tamsin L. Edwards, Jonathan M. Gregory, George Koubbe, Anders Levermann, Andre Merzky, Sophie Nowicki, Matthew D. Palmer, and Chris Smith

<https://doi.org/10.5194/gmd-16-7461-2023>

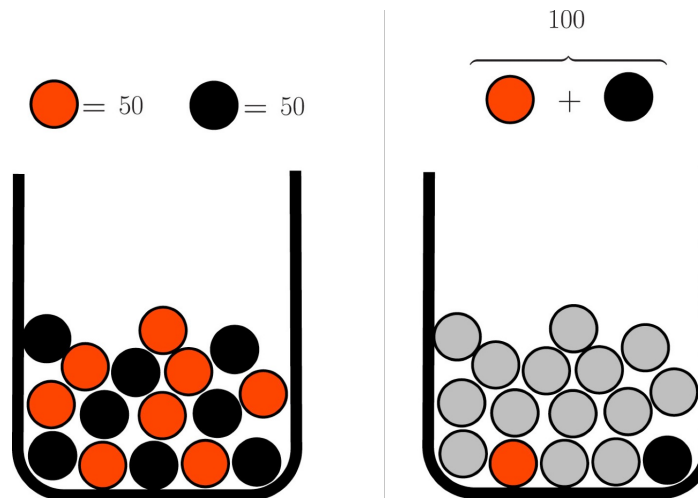
Section 1 :

Deep Uncertainty, Ambiguity & FACTS

Ambiguity is a form of deep uncertainty that is difficult to quantify and incorporate into traditional decision-making frameworks

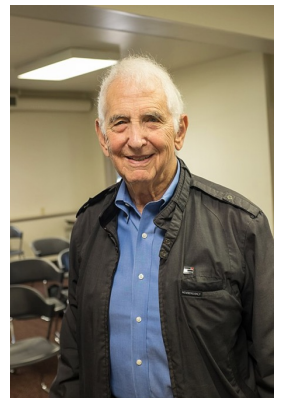
Quantifiable Uncertainty

Can be expressed with a well-defined probability distribution.



Ambiguity

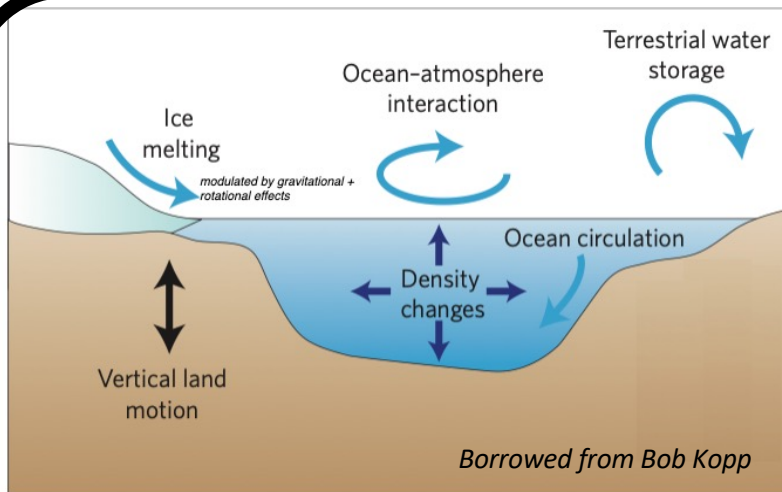
Cannot be represented by a single probability distribution



Daniel Ellsberg
(Source: Wikipedia)

Types of Uncertainty

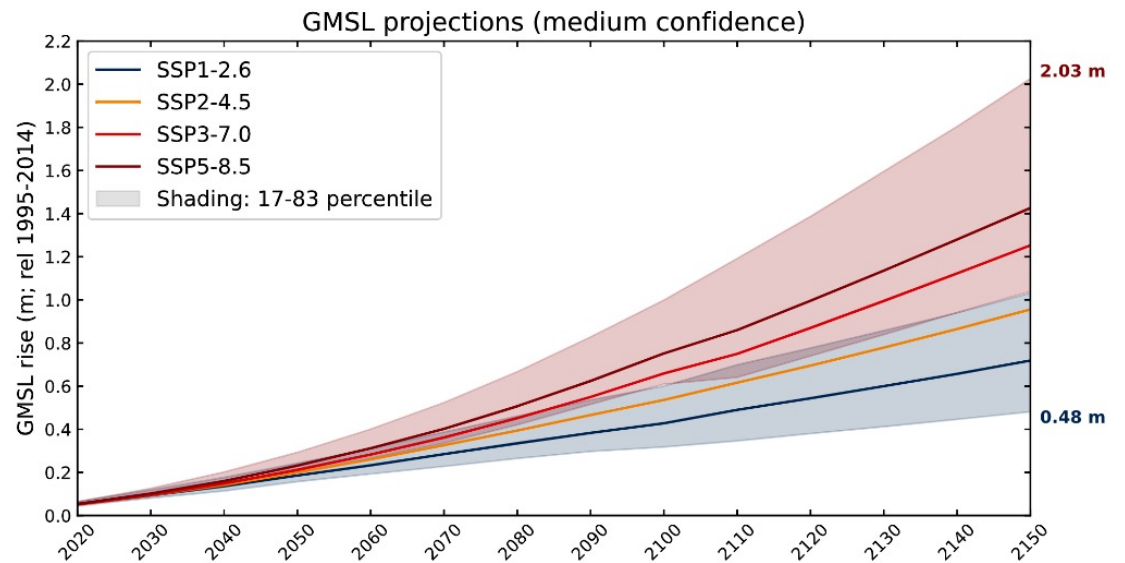
Borrowed from Bob Kopp



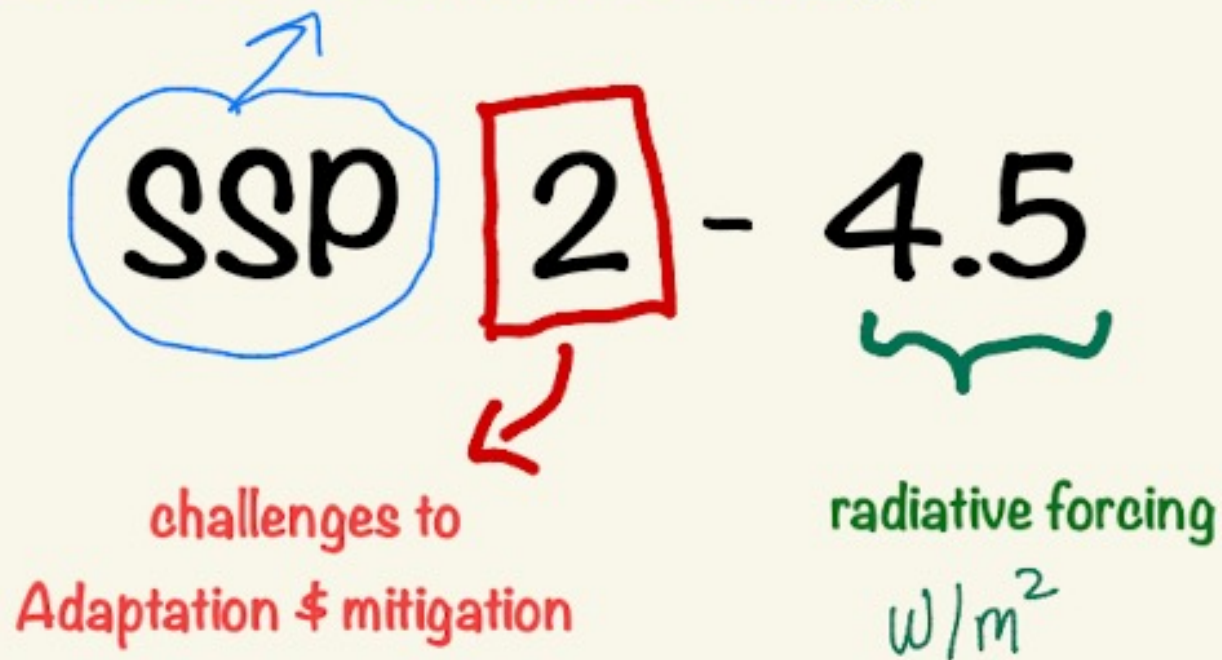
Very simplified schematic of processes influencing sea-level change

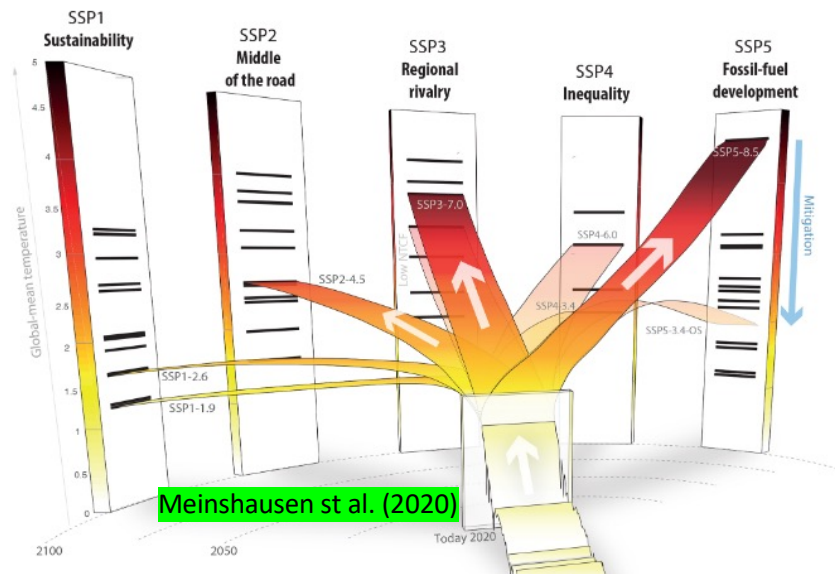
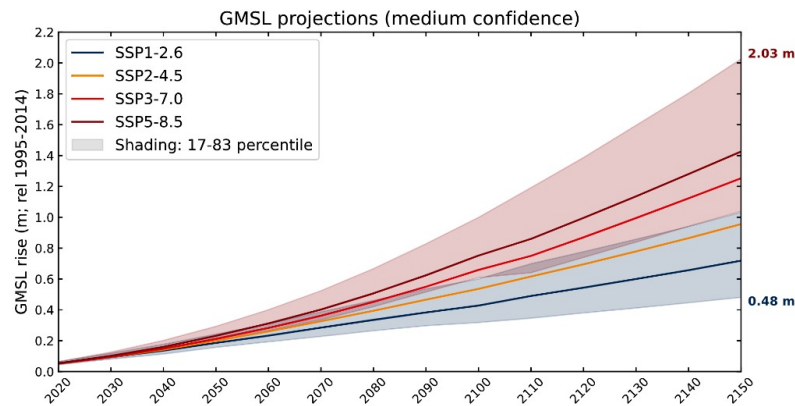
Sea-level projections integrate multiple data sources, but significant **uncertainties**-like **within ice-sheet processes** and **socioeconomic and policy dynamics**—complicate accurate risk assessment.

Probabilistic Sea-level Projections



shared socioeconomic Pathways





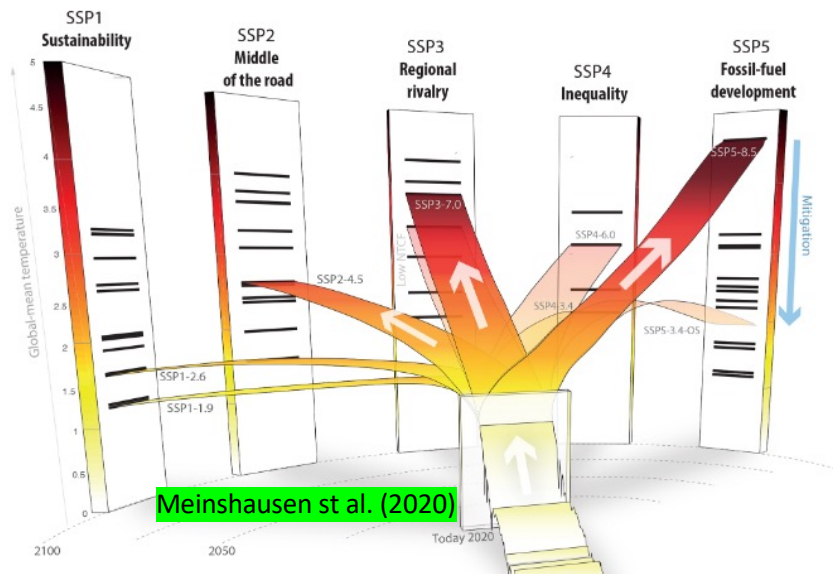
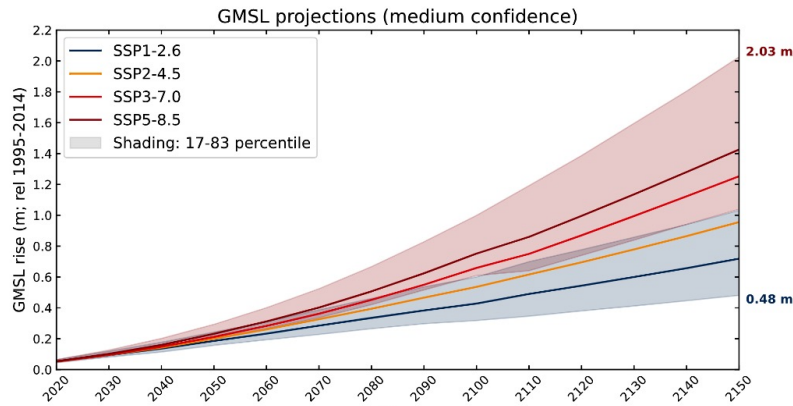
shared socioeconomic pathway (SSP)

- SSP1 – Sustainability :
 - Eco-focused, equitable development
 - Low challenges to mitigation & adaptation

- SSP2 – Middle of the Road:
 - Continuation of current trends
 - Moderate challenges to both

- SSP3 – Regional Rivalry :
 - Fragmented, nationalistic world
 - High challenges to both

- SSP5 – Fossil-fueled Development :
 - Rapid growth via fossil fuels
 - High mitigation, low adaptation challenges



Representative Concentration Pathways (RCPs)

RCP2.6: A low emissions scenario aiming to limit global warming to below 2°C above pre-industrial levels.

RCP4.5: A stabilization scenario where radiative forcing stabilizes at 4.5 W/m² by 2100.

RCP 7.0: is a baseline, not a mitigation target, it reflects medium to high emissions and warming without new climate policies.

RCP8.5: A high emissions scenario with continued increases in greenhouse gas emissions causing radiative forcing of 8.5 W/m² by 2100.

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(praveen.kumar@Rutgers.edu)

Representative Concentration Pathways (RCPs)

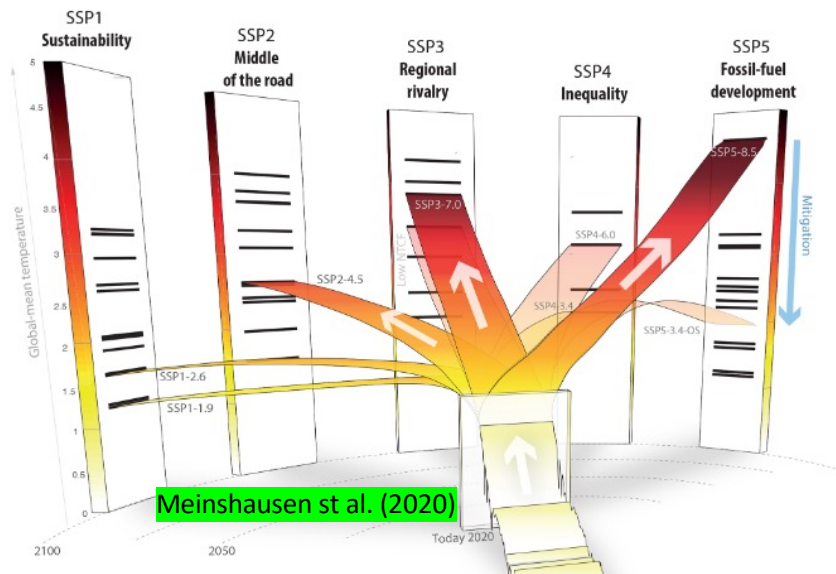
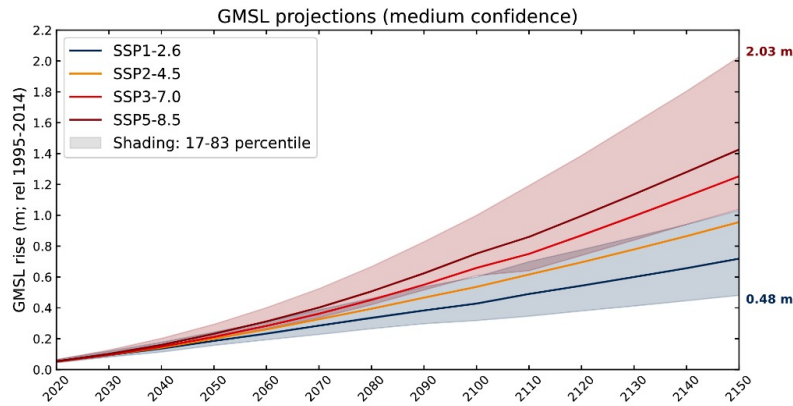
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Integration of SSP and RCP



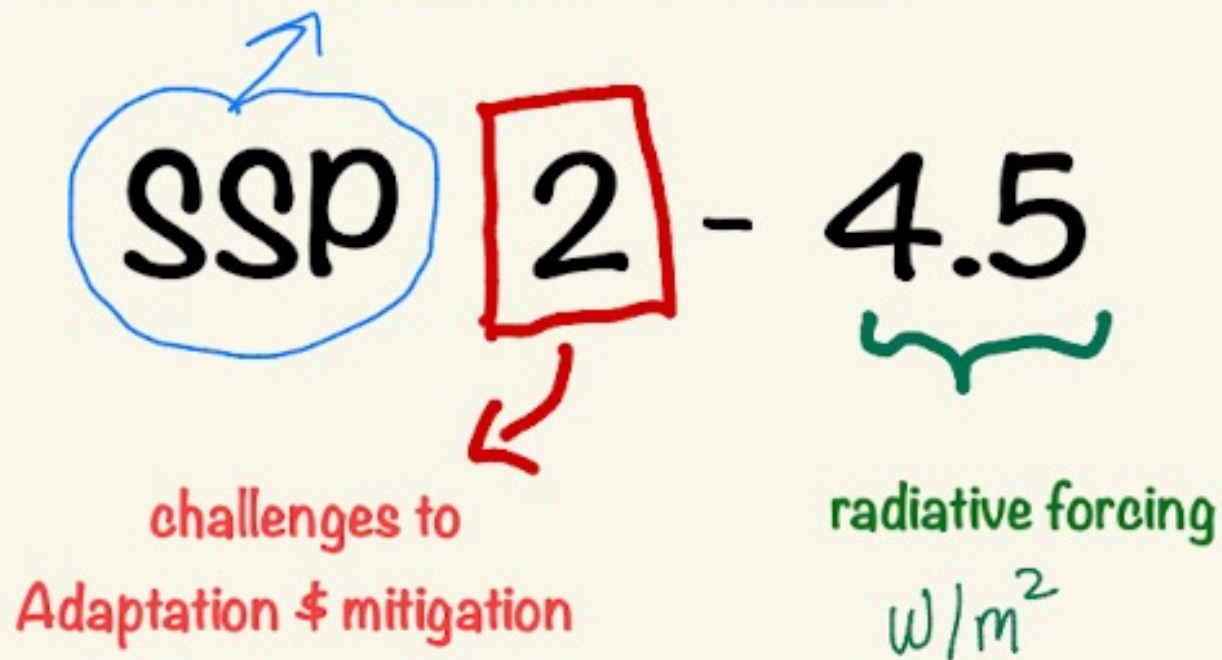
SSP1-2.6 assumes a world making progress toward sustainability, low challenges to mitigation and adaptation but with slightly higher greenhouse gas emissions.

SSP2-4.5 assumes a future where trends broadly follow historical patterns, with medium challenges to mitigation and adaptation.

SSP3-7.0 envisions a fragmented world, characterized by high challenges to both mitigation and adaptation.

SSP5-8.5 assumes a world with high economic growth driven by fossil fuels, resulting in high challenges to mitigation.

shared socioeconomic Pathways



AR7 2028 will have new Scenarios,

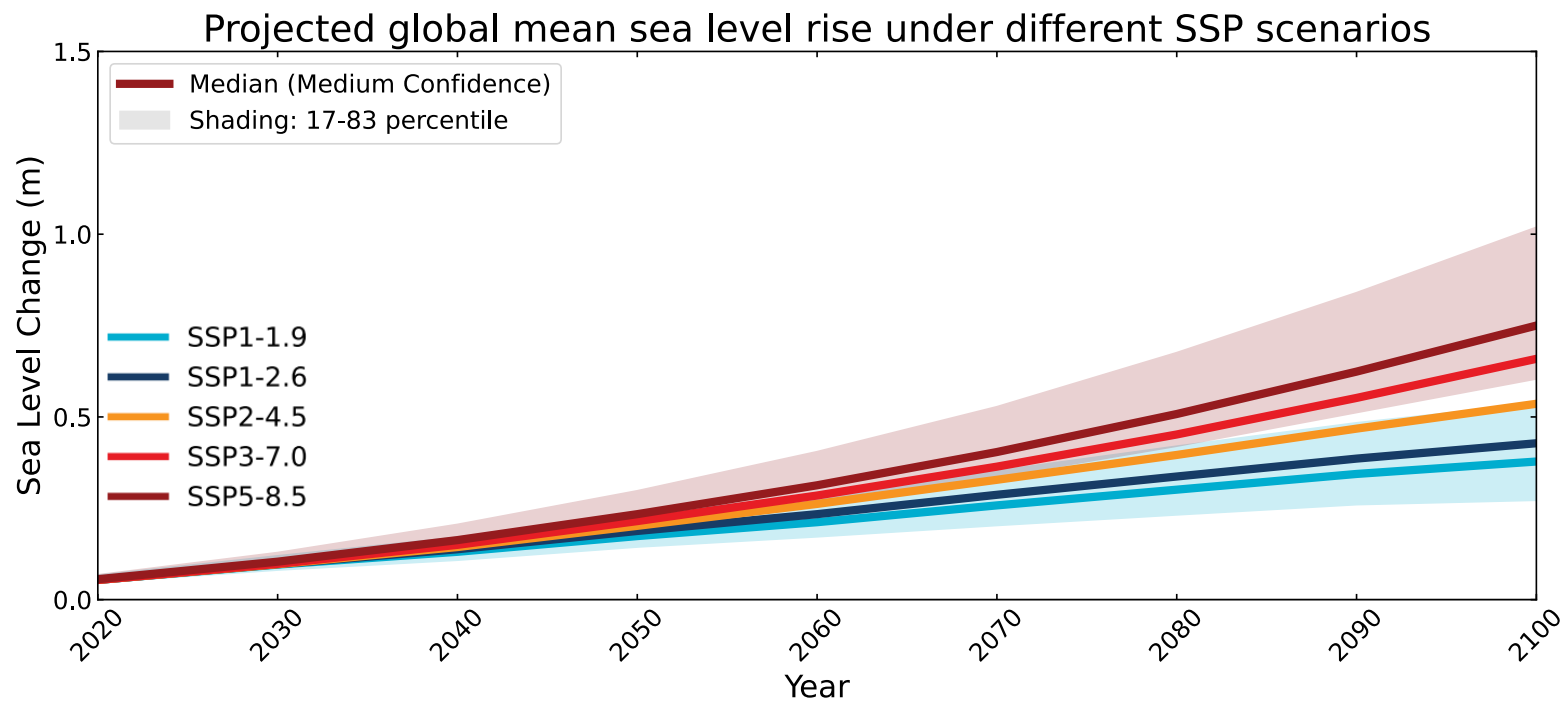
We are currently working on projections
stay tuned ...

Key contributors to ambiguity in sea-level rise projections

- ❖ **Socioeconomic Factors:** especially those influencing future emissions

Limitations

- Uncertain Future Emissions

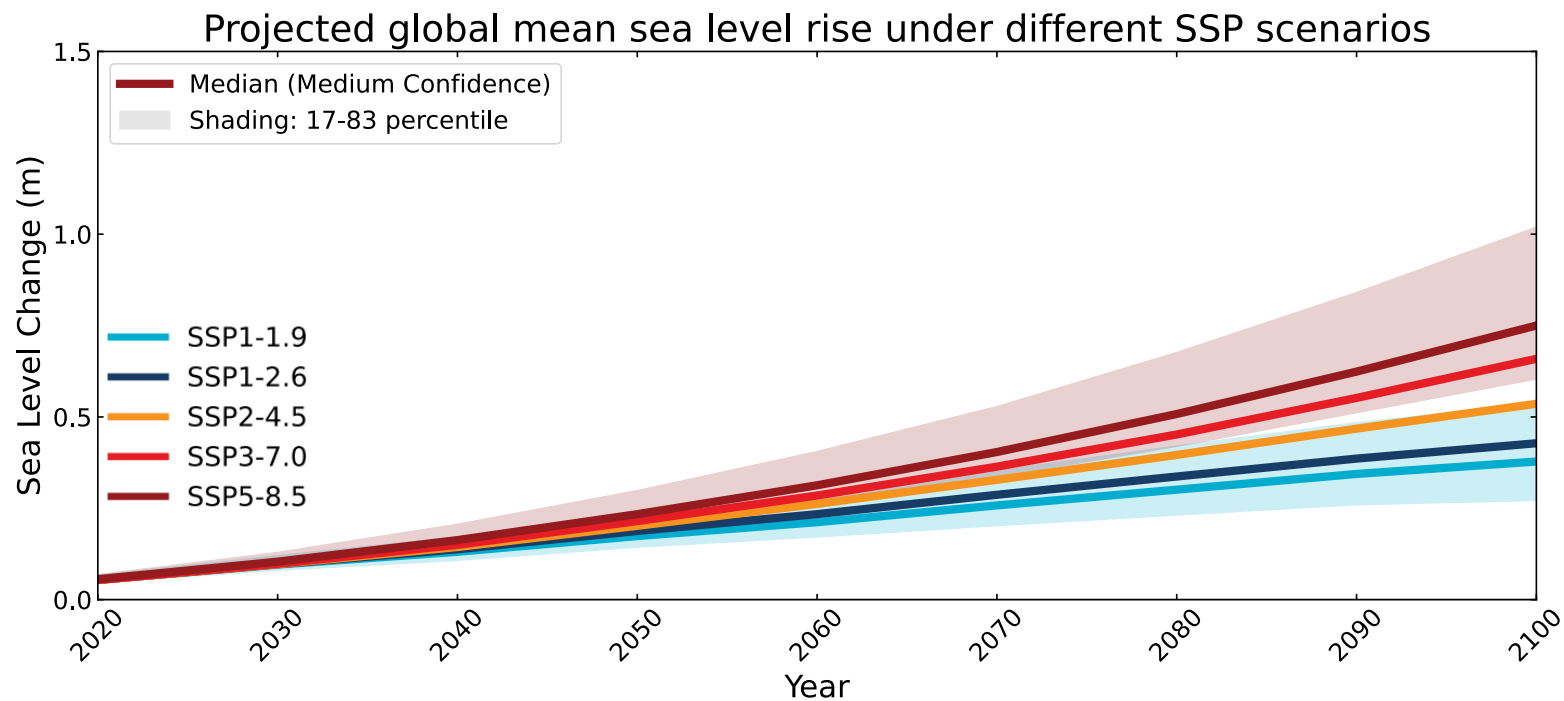


KUDOS MBL



Key contributors to ambiguity in sea-level rise projections

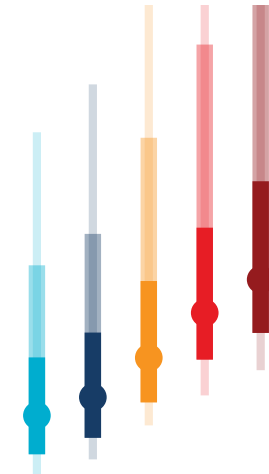
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Key contributors to ambiguity in sea-level rise projections

- ❖ **Socioeconomic Factors:** especially those influencing future emissions
- ✓ **Adding in unknown-likelihood, high impact events (MICI/MISI)**

- ✓ Adding in unknown-likelihood, high impact events (MICI/MISI)



For an in-depth sea-level Uncertainty review paper

Perspective | Published: 19 September 2025

Progress and future directions in constraining uncertainties in sea-level projections using observations

[Denis Felikson](#) , [David R. Rounce](#), [John Fasullo](#), [Angelica Rodriguez](#), [Surendra Adhikari](#), [Brett Buzzanga](#), [Sönke Dangendorf](#), [Robert E. Kopp](#), [Richard B. Lammers](#), [J. T. Reager](#), [Doug Brinkerhoff](#), [Beata Csatho](#), [Manuela Girotto](#), [Benjamin Hamlington](#), [Erik R. Ivins](#), [Praveen Kumar](#), [Eric Larour](#), [R. Steven Nerem](#), [Sophie Nowicki](#), [Nicole-Jeanne Schlegel](#), [Jan-Erik Tesdal](#) & [Matthew Weathers](#)

[Nature Climate Change](#) **15**, 1039–1051 (2025) | [Cite this article](#)



<https://www.nature.com/articles/s41558-025-02437-4>

Section 2 : Uncertainty Communication

IPCC assessment reports

To communicate effectively, you need an audience;
the larger the audience, the more people receive your message.

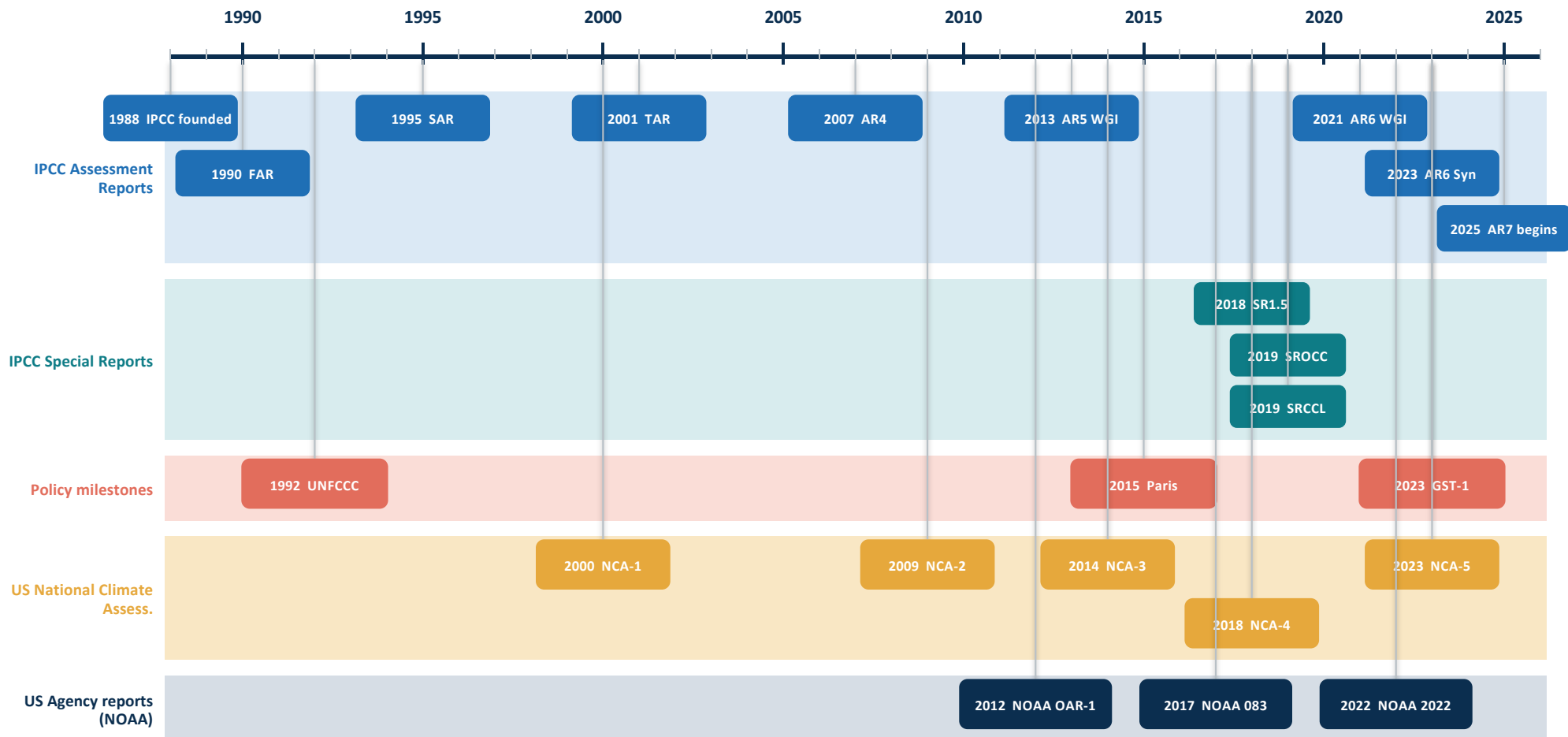
... Therefore, reaching a wider audience is essential.
National and international reports help achieve this by bringing
together global experts.

What is the IPCC?

The Intergovernmental Panel on Climate Change

- A UN body founded in 1988.
- 195 governments are members.
- Rather than conducting new research, it assesses global climate literature to compile consensus reports.

IPCC Timeline 1988 – 2025



Section 3 : Tool to

Assess & Communicate Uncertainty

Framework for Assessing Changes To Sea-level (FACTS)

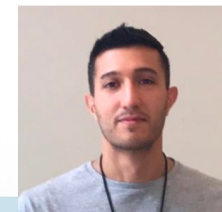
slide courtesy @ [Kelly McCusker](#)

The FACTS community

FACTS has a core development team and members of the science community who contribute in various ways

Core team

- Robert Kopp (Dept of Earth & Planetary Sciences; Rutgers), Shantenu Jha (PI RADICAL Lab; Rutgers): PIs and founders of FACTS
- FACTS development team
 - Kelly McCusker (Associate Director, Rhodium Group, Climate Impact Lab): oversees FACTS1 maintenance and FACTS2 development
 - Emma Marshall (Research Software Developer, Rhodium Group, Climate Impact Lab): primary FACTS2 software developer
 - Brewster Malevich (Senior Research Software Developer, Rhodium Group, Climate Impact Lab) advises on FACTS2 design and implementation
 - Praveen Kumar (postdoc, Rutgers, Climate Impact Lab): sea level researcher, FACTS power user, and contributor to FACTS1
 - Matteo Turilli (RADICAL Lab, Dept of Electrical & Computer Engineering; Rutgers): advises on software design, RADICAL runtime tools
 - Aymen Al Saadi (RADICAL Lab; Rutgers): advises on software design, RADICAL runtime tools



RHODIUM GROUP

The FACTS community

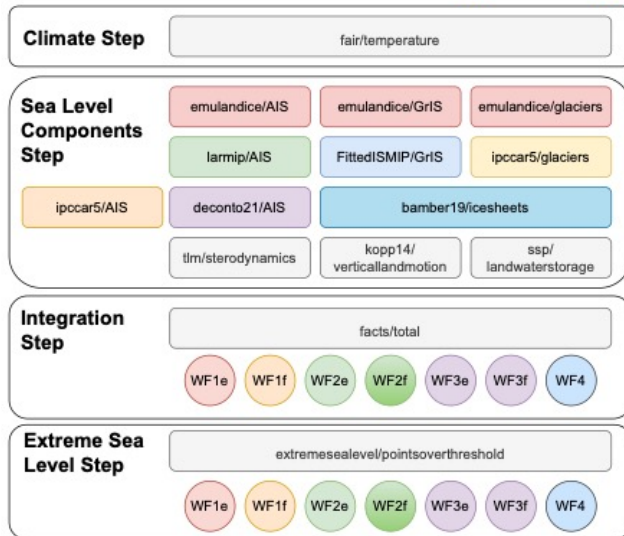
FACTS has a core development team and members of the science community who contribute in various ways

Selection of the wider science community

- Tamsin Edwards (Kings College London): land ice emulation (emulandice2)
- Aimée Slangen (NIOZ Royal Netherlands Institute for Sea Research): regional GRD fingerprinting
- Victor Malagón-Santos (TU Delft): sterodynamics, GIA/VLM
- Tim Hermans (Utrecht Univ): extreme sea level
- Julius Oelsmann (Tulane Univ): VLM
- Denis Felikson (NASA Goddard): upcoming ISMIP7
- Former FACTS developers
 - Greg Garner
 - Alex Reedy



Framework for Assessing Changes To Sea-level (FACTS)

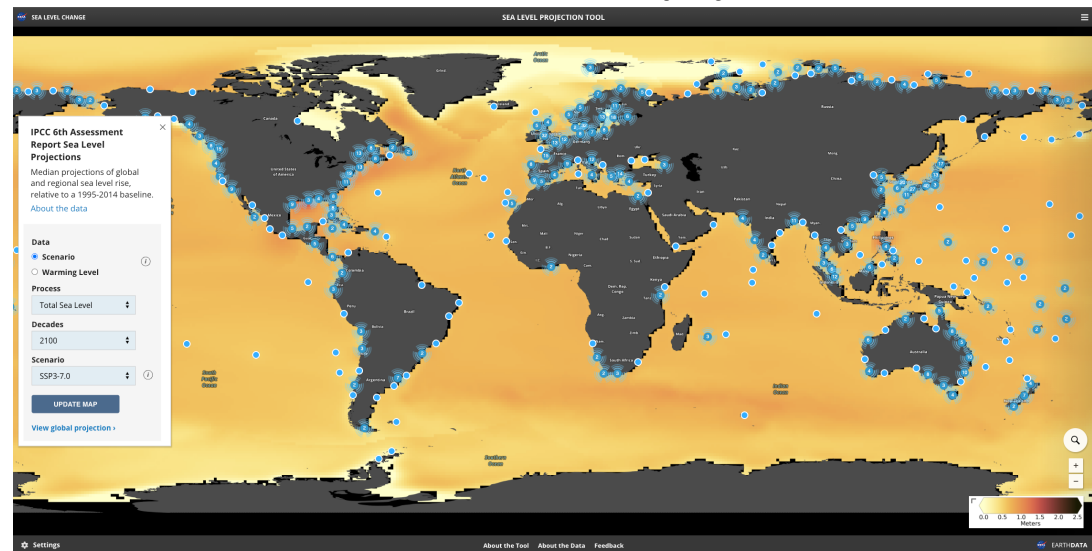


GMD: Kopp et al, 2023



<https://github.com/radical-collaboration/facts>

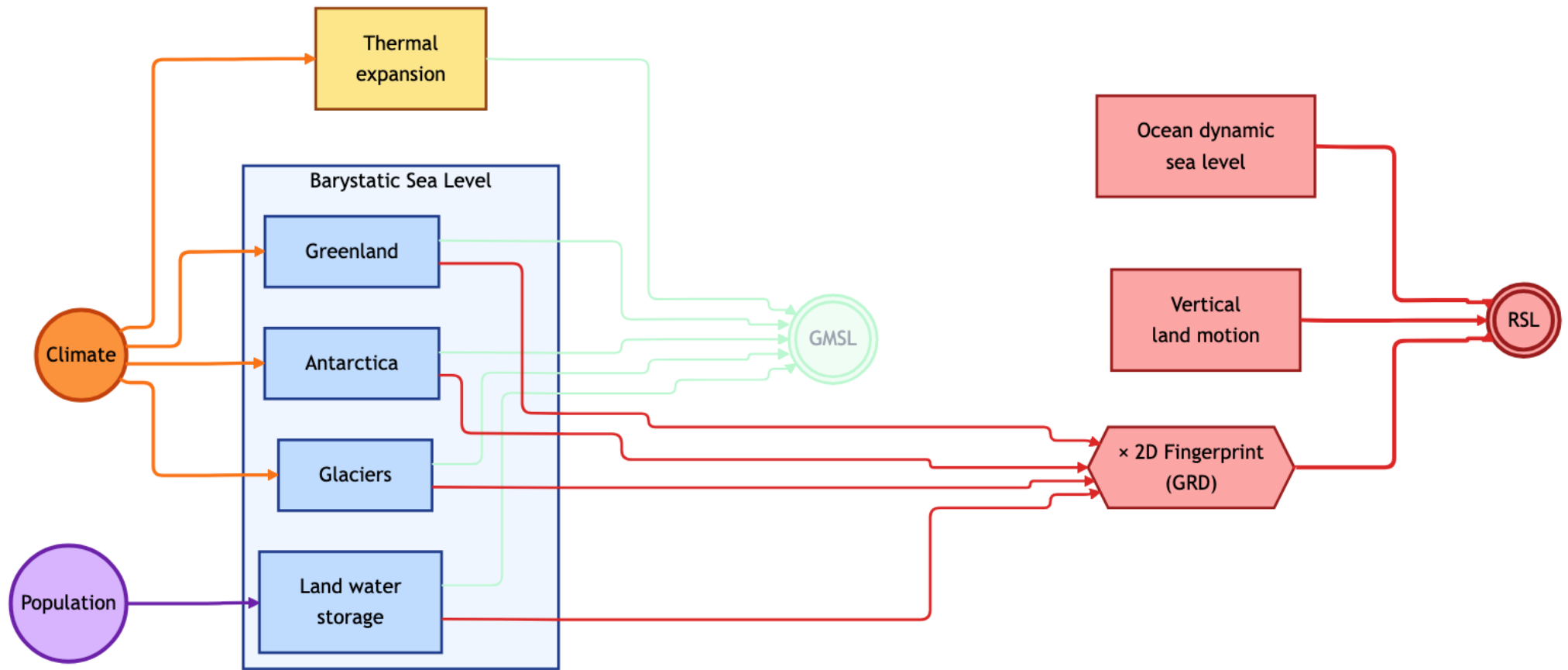
NASA IPCC AR6 sea-level-projection-tool



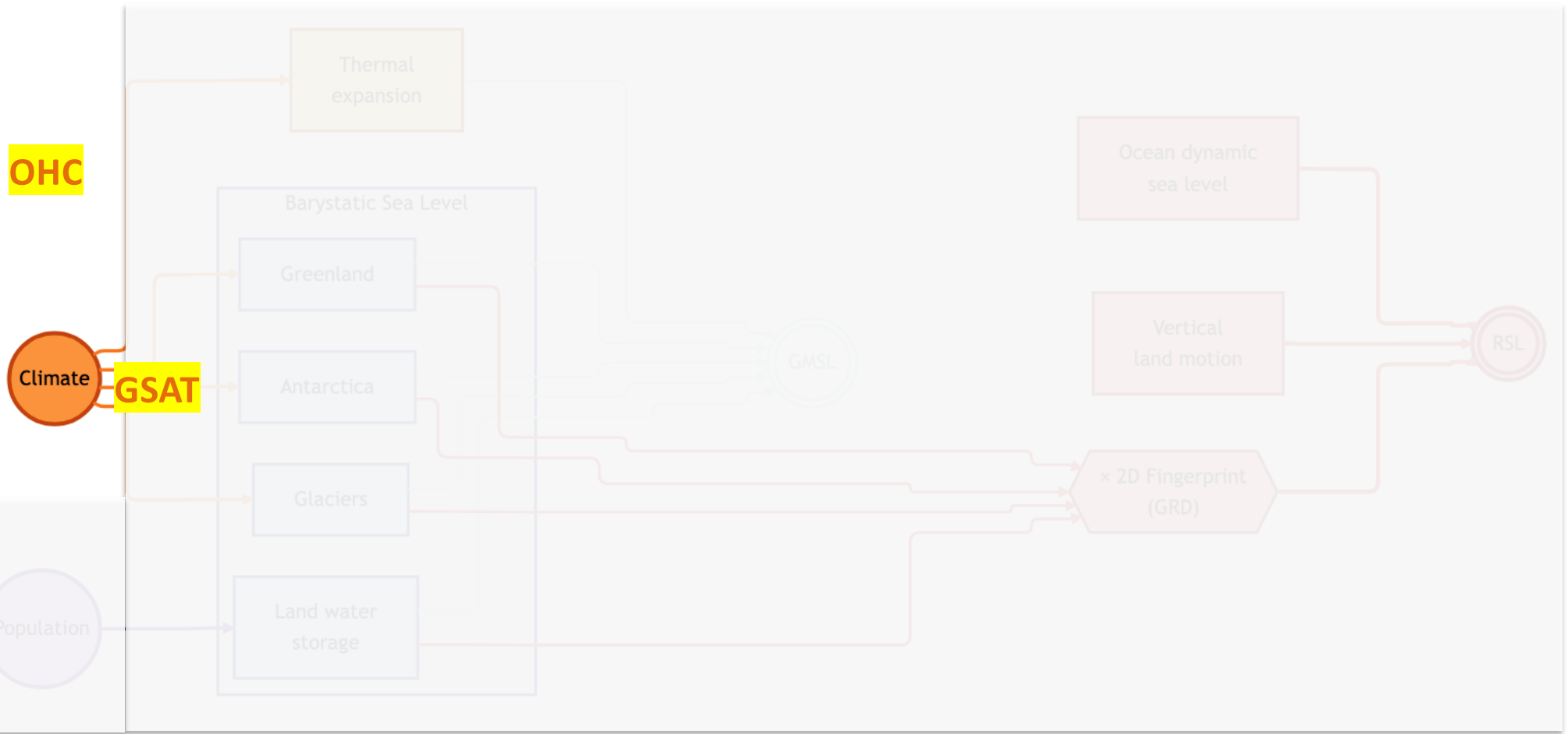
<https://sealevel.nasa.gov/ipcc-ar6-sea-level-projection-tool>



Framework Design



Framework Design



Emulators

*Simplified models that replicate key behaviors of more complex Models
... **but run significantly faster.***



Speed

Run hundreds of scenarios in minutes instead of months



Accessibility

Runs on personal computers



Integration

Easily incorporated into other models

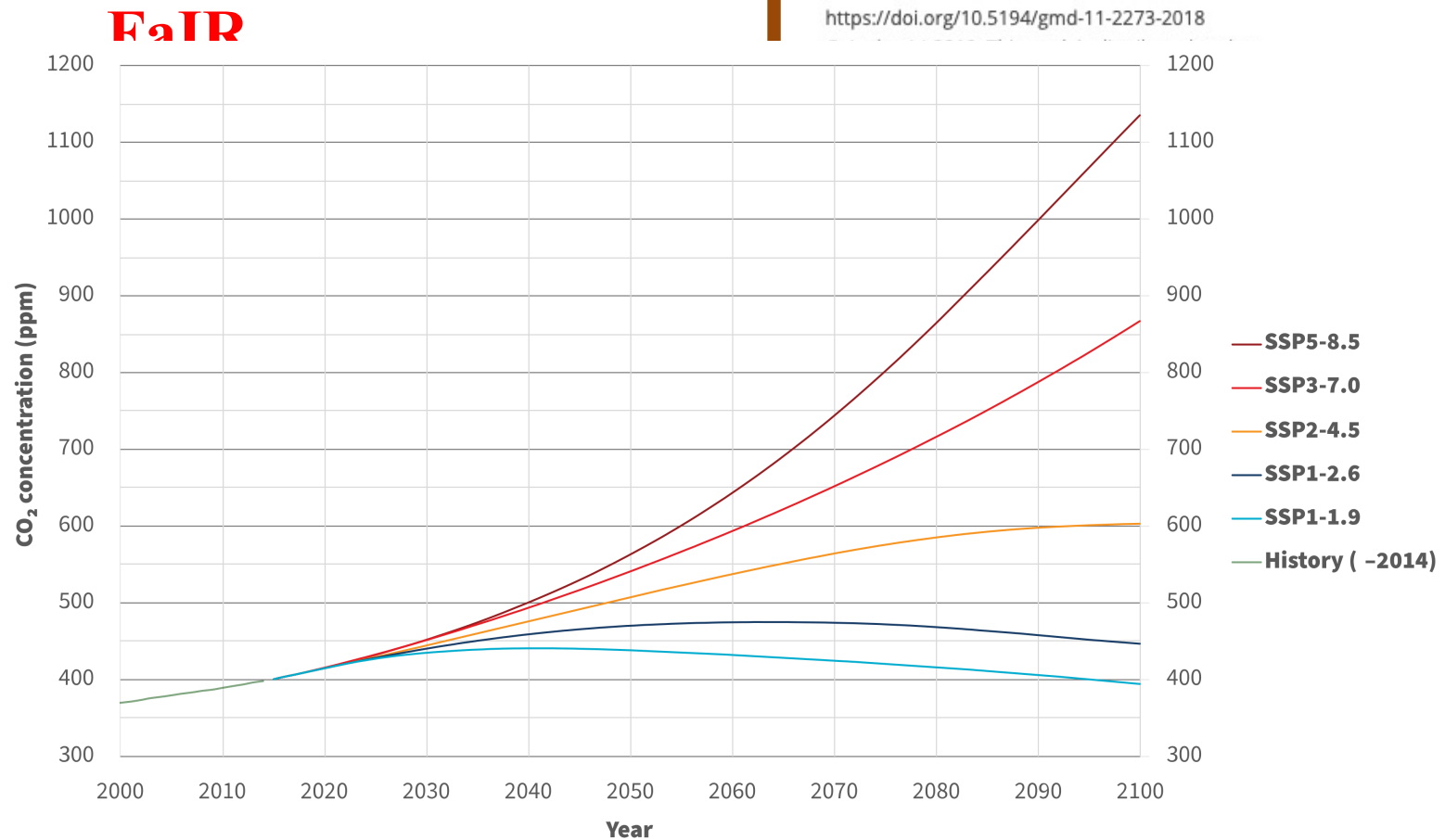
Bridging the gap between complex Earth System Models and rapid climate policy assessment needs

Disadvantage

- Simplified representation of processes (e.g., regional climate feedbacks)
- Parameter uncertainty in carbon cycle and climate response
- Requires careful calibration and validation

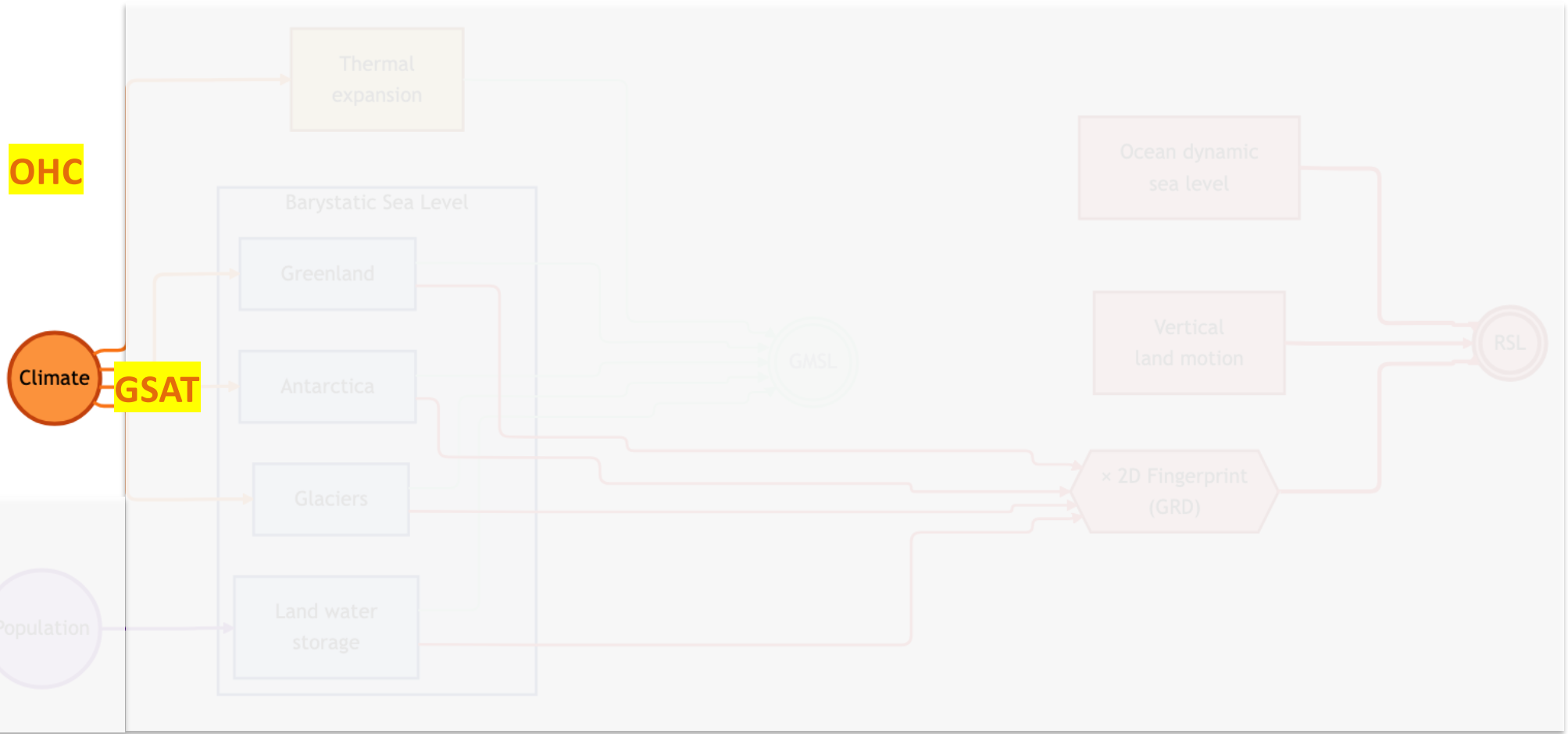
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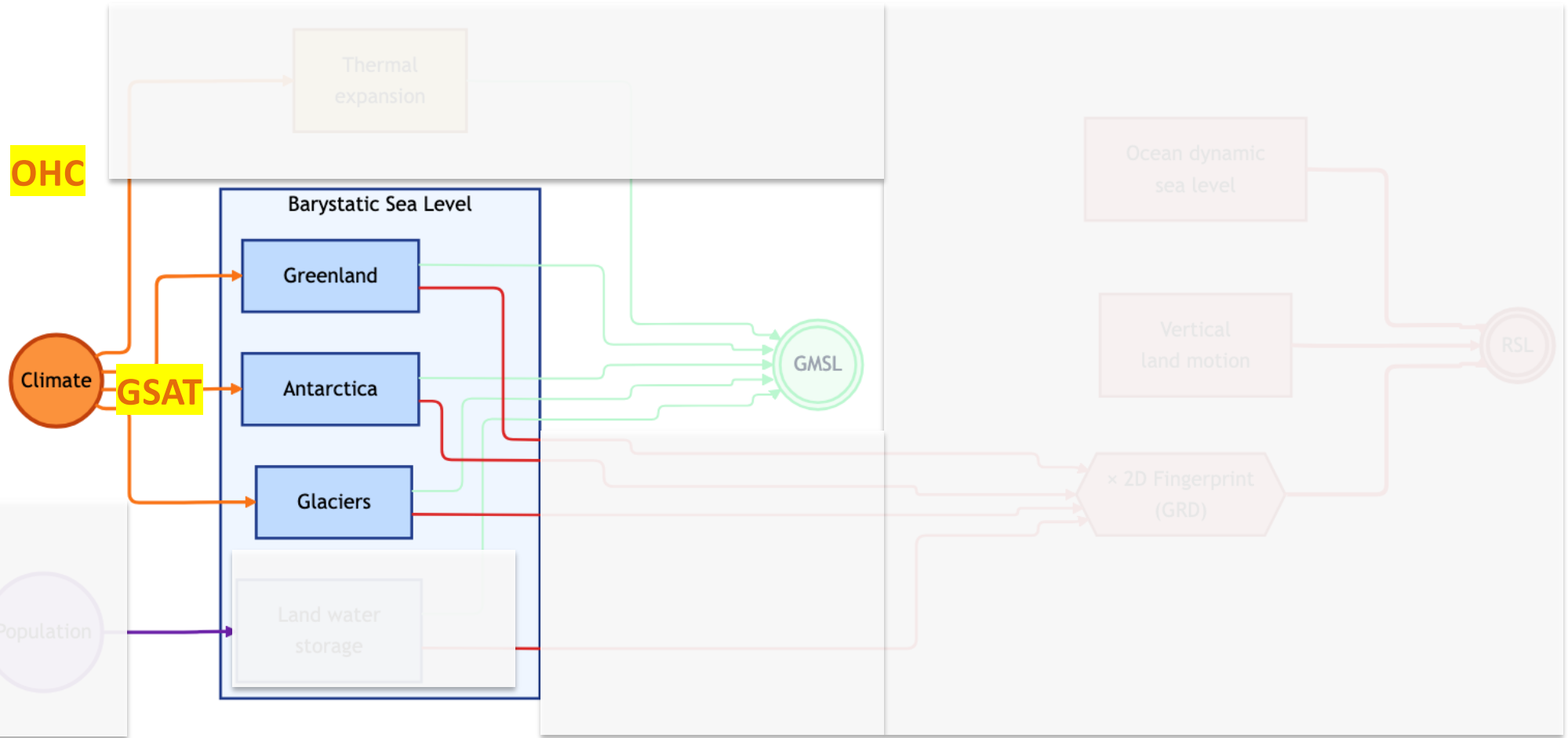


- Inputs (**I use**) : Greenhouse gas Emissions (CO₂ , CH₄ , N₂O)
- ✓ Outputs: global mean temperature and ocean heat content.

Framework Design

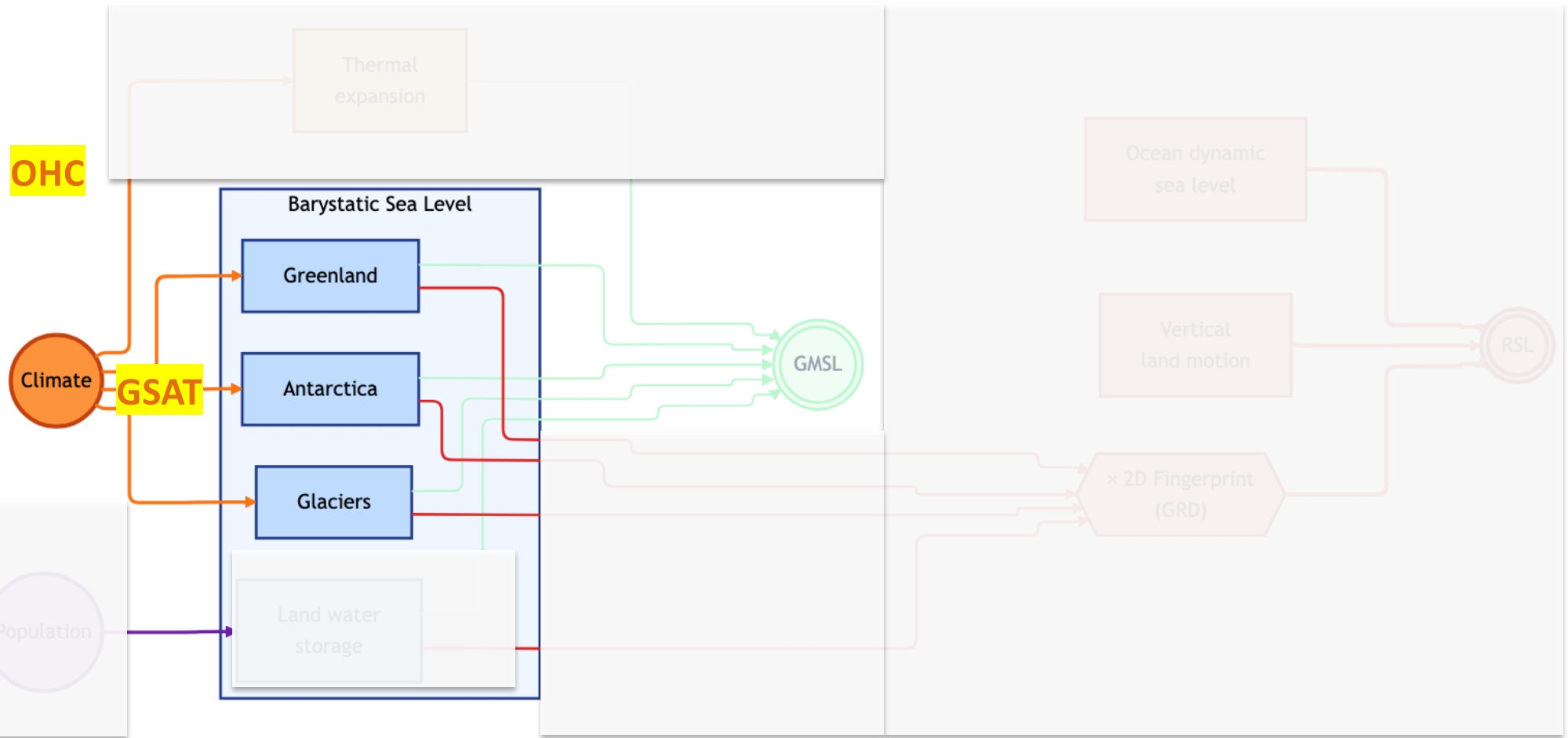


Framework Design

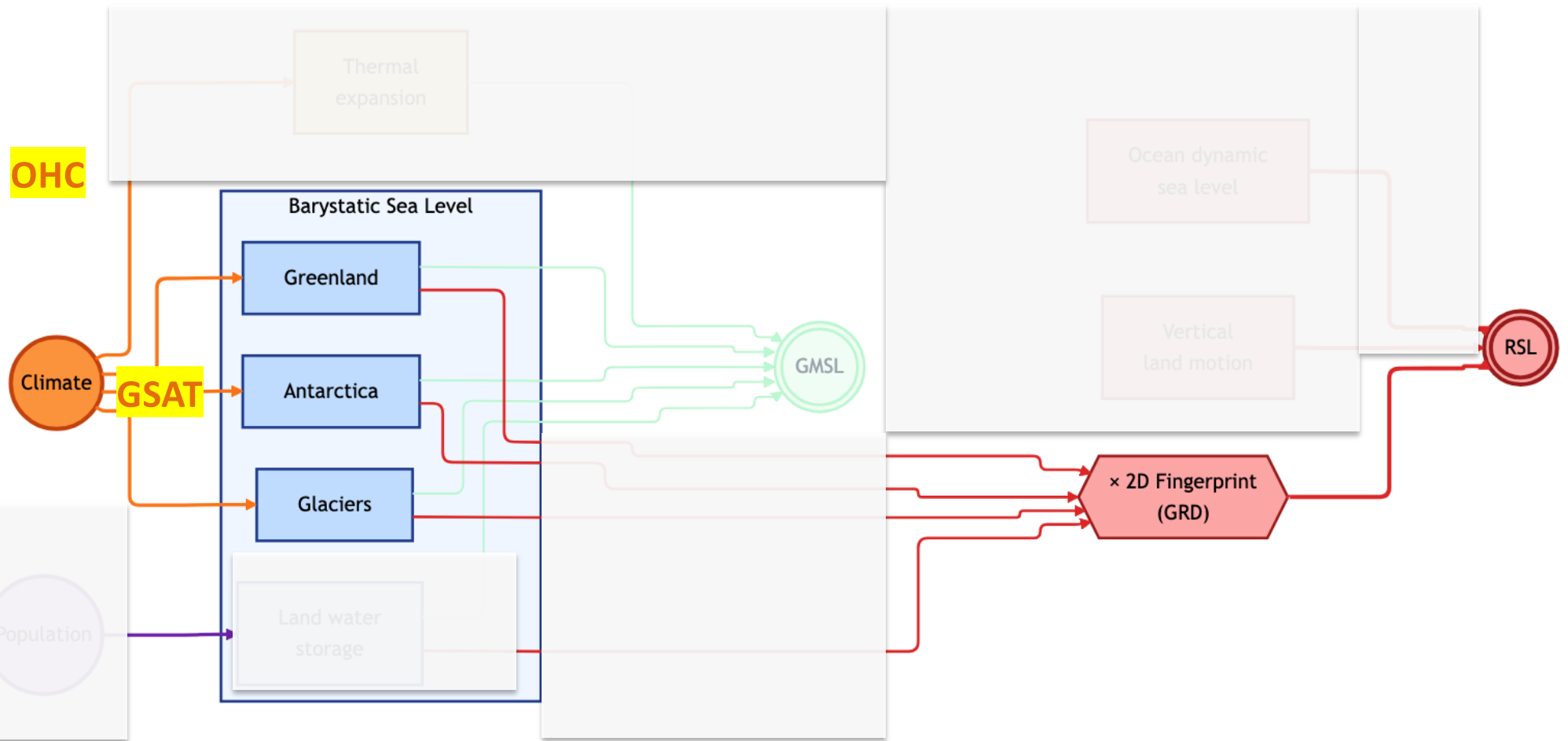




Framework Design



Framework Design

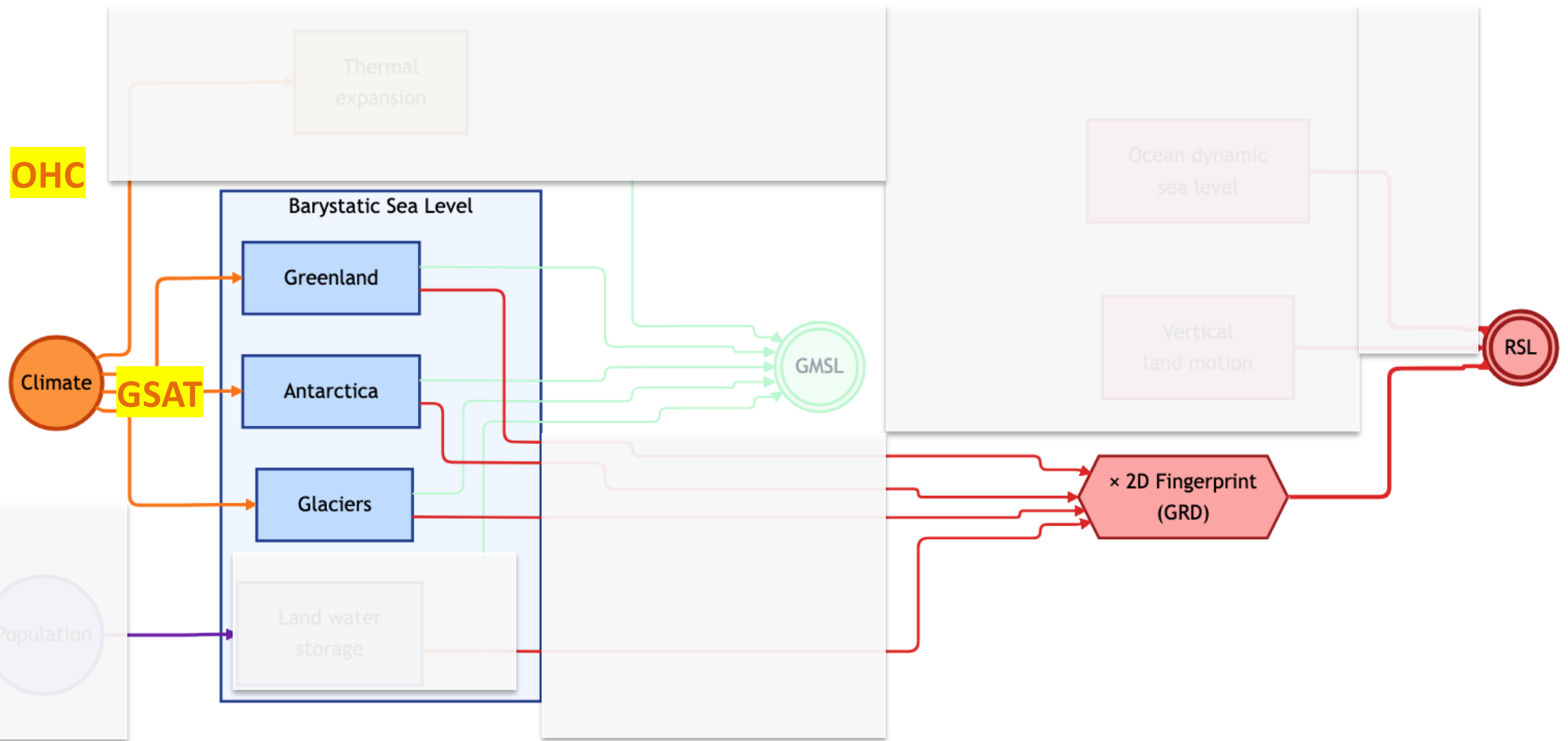


Sea-Level Fingerprint

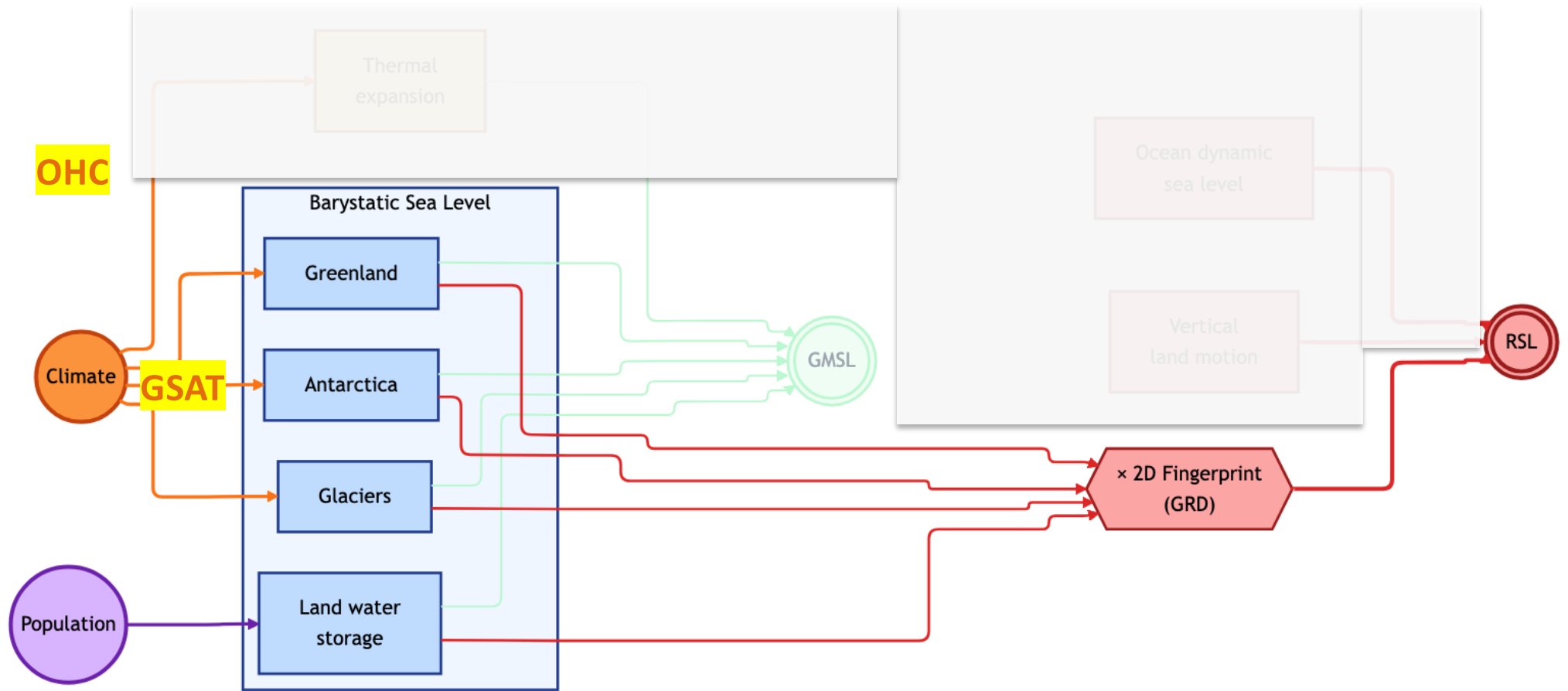
- When ice melts, sea level does NOT rise uniformly around the globe
 - ✓ nearby ocean actually FALLS
 - ✓ Far-field ocean rises more than the global mean to compensate
- This spatial pattern = the 'fingerprint' of that ice source
 - ✓ Also includes rotational feedback: shifting ice mass changes Earth's rotation axis

Gravitational + Rotational + Deformational (GRD) fingerprint

Framework Design



Framework Design

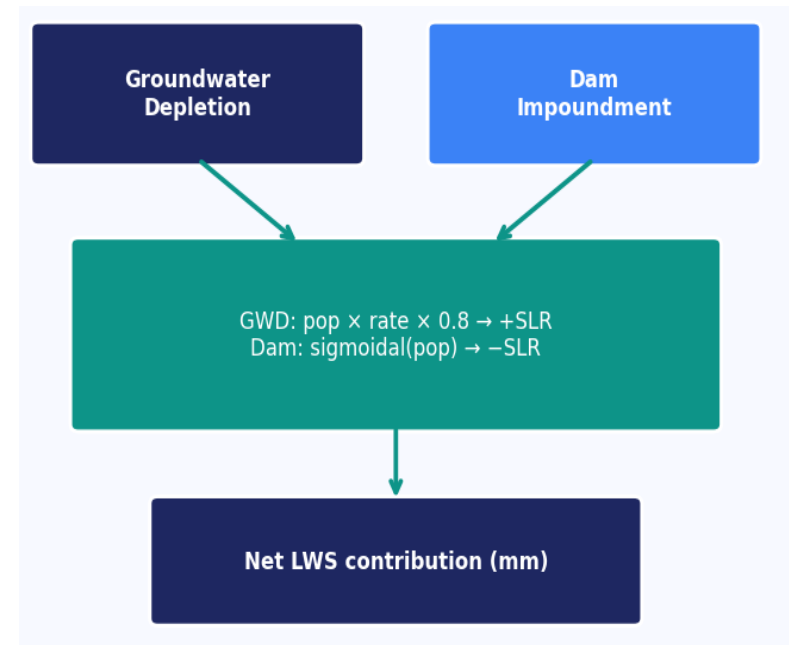


Land Water Storage

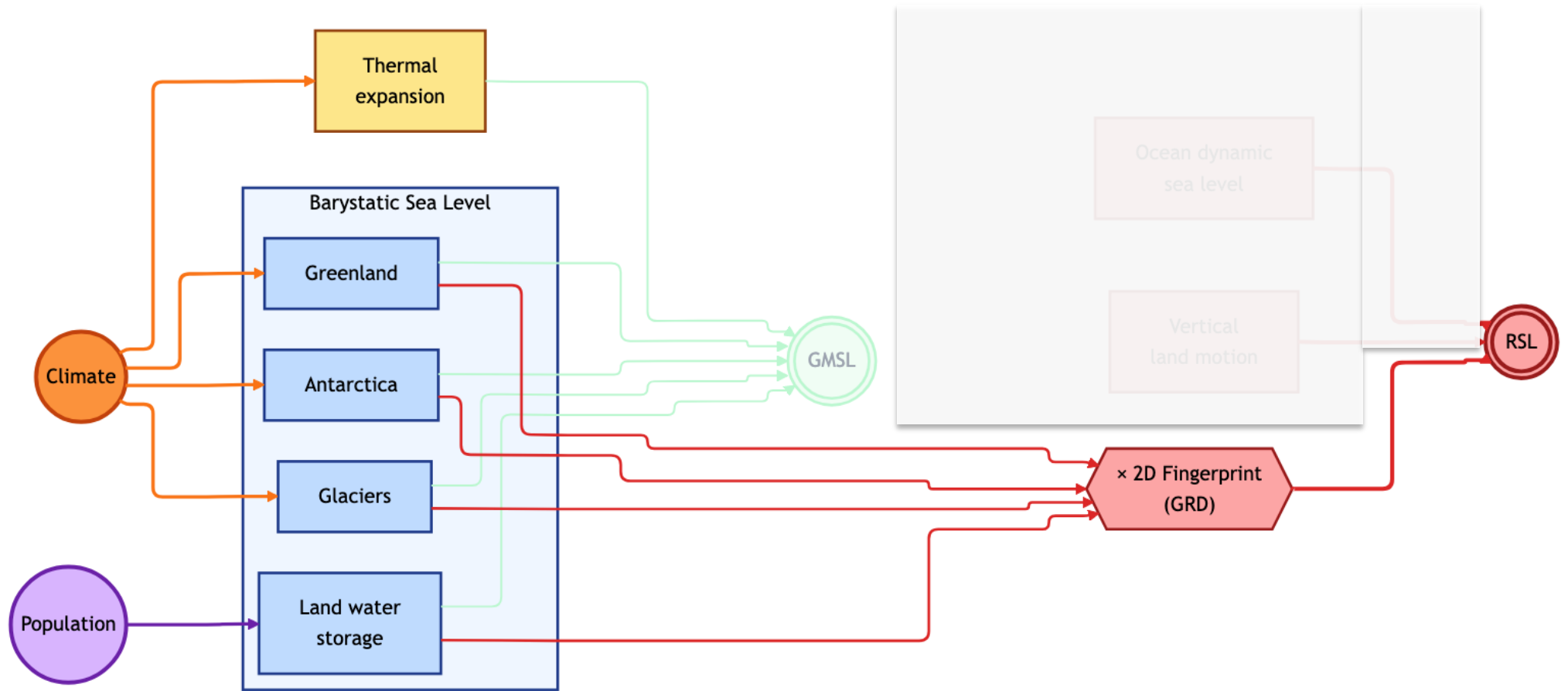
Two competing processes:

- **Groundwater depletion** (+SL) : pumping exceeds recharge
 - 80% of depleted groundwater reaches the ocean (Wada et al. (2016))
- **Dam impoundment** (-SL) : reservoirs trap water on land, reducing ocean volume
 - 2020–2040 (infrastructure planning lag)
- Net effect: post-2000 groundwater depletion increasingly dominates (dam building slowed)

After ~2040: GWD rate increases; dam impoundment saturates
→ net positive LWS contribution



Framework Design

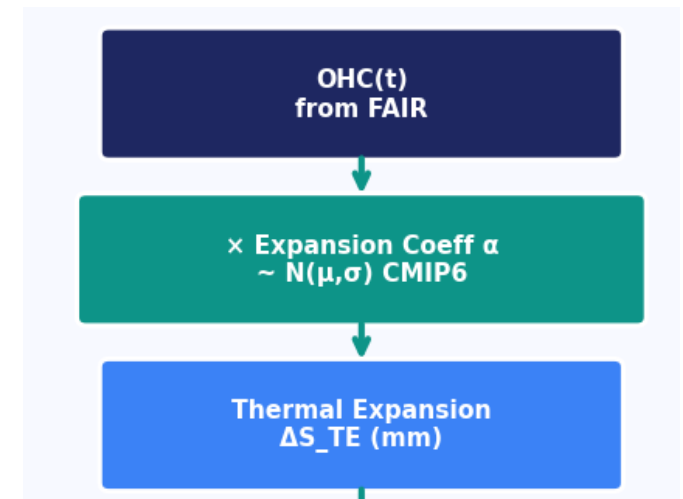


Sterodynamic Sea Level Change

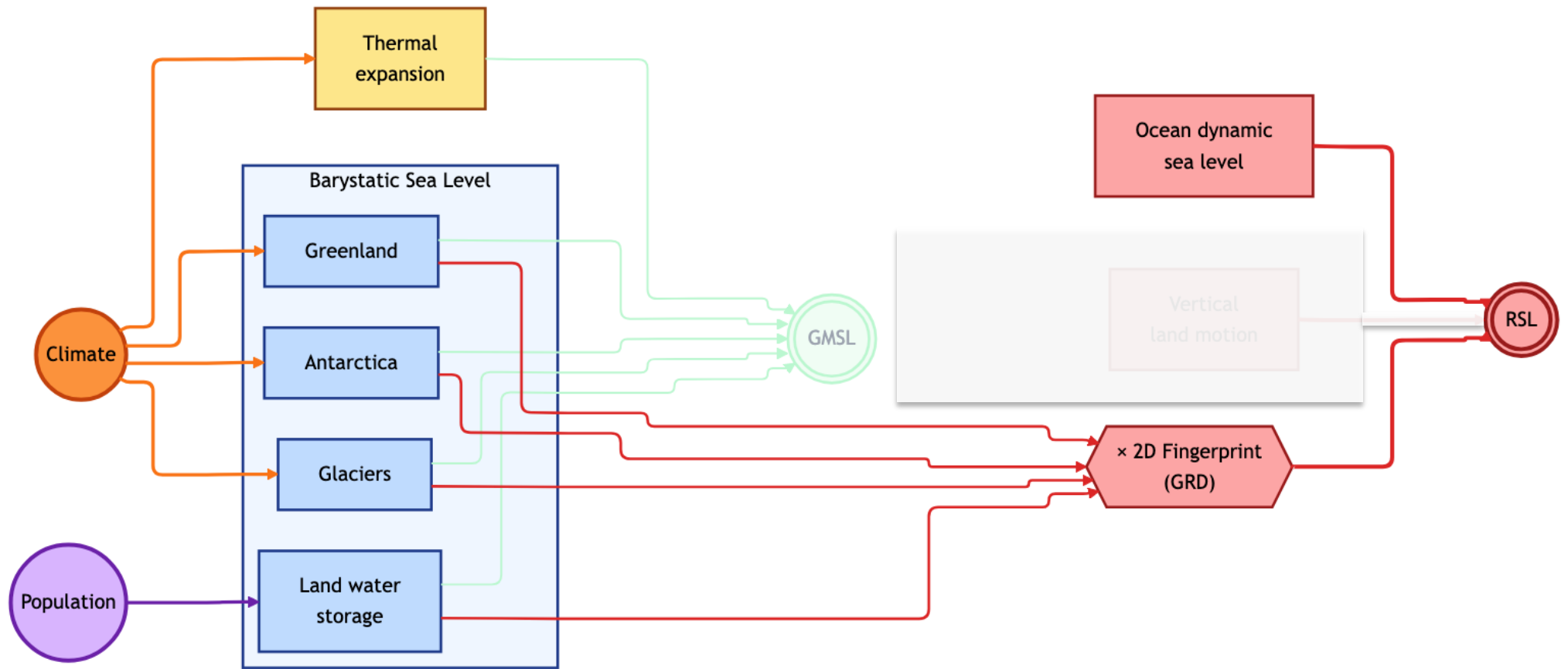
(1) Thermosteric SL

multiplies OHC samples (from FAIR) X expansion coefficient (mm)

Expansion coefficient α calibrated to 18 CMIP6 models



Framework Design



Sterodynamic Sea Level Change

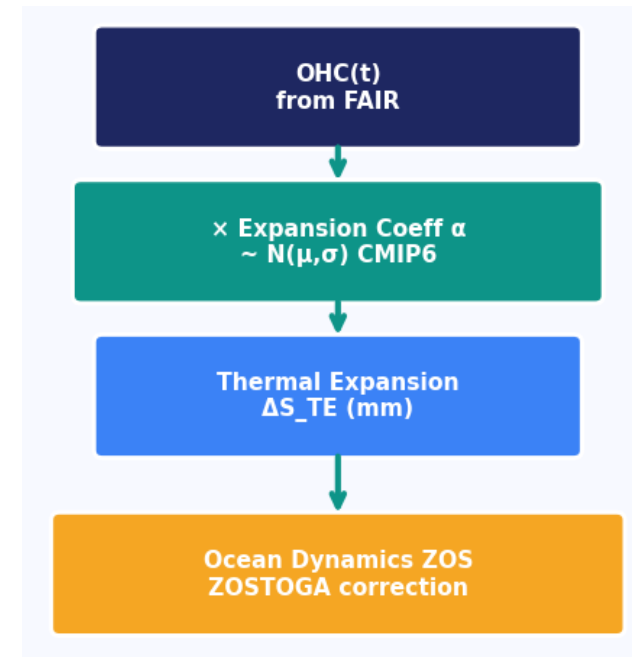
(1) Thermal expansion from ocean warming

Multiplies OHC samples (from FAIR) by an expansion coefficient (that's how you get mm of sea-level rise)

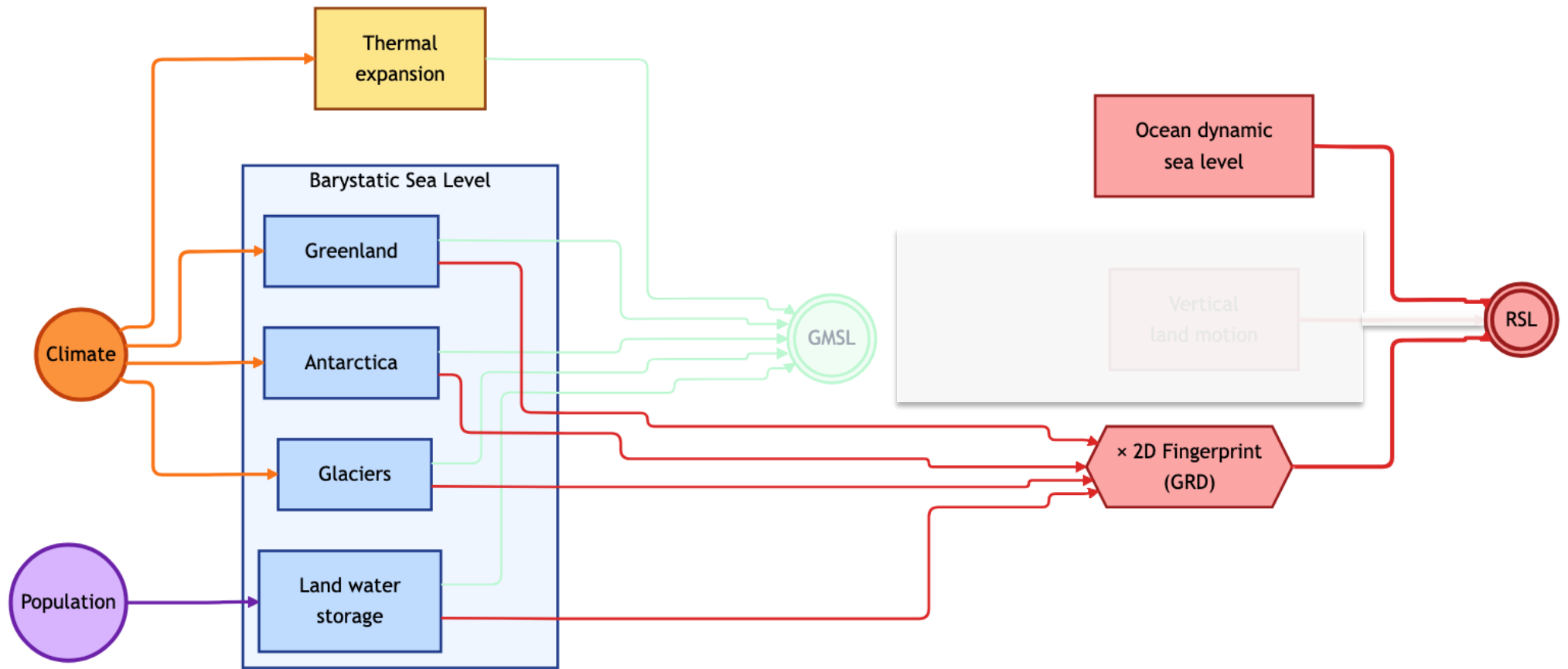
Expansion coefficient α calibrated to 18 CMIP6 models

(2) Dynamic sea level from circulation changes.

- Ocean dynamics (ZOS): uses CMIP6 ZOS (dynamic sea surface height) fields for localization
- Global mean thermosteric component is subtracted before spatial interpolation to avoid double-counting



Framework Design



The VLM in the Kopp et al. (2014) framework is a hybrid method: it uses mathematical **modeling** to analyze **observed** data.

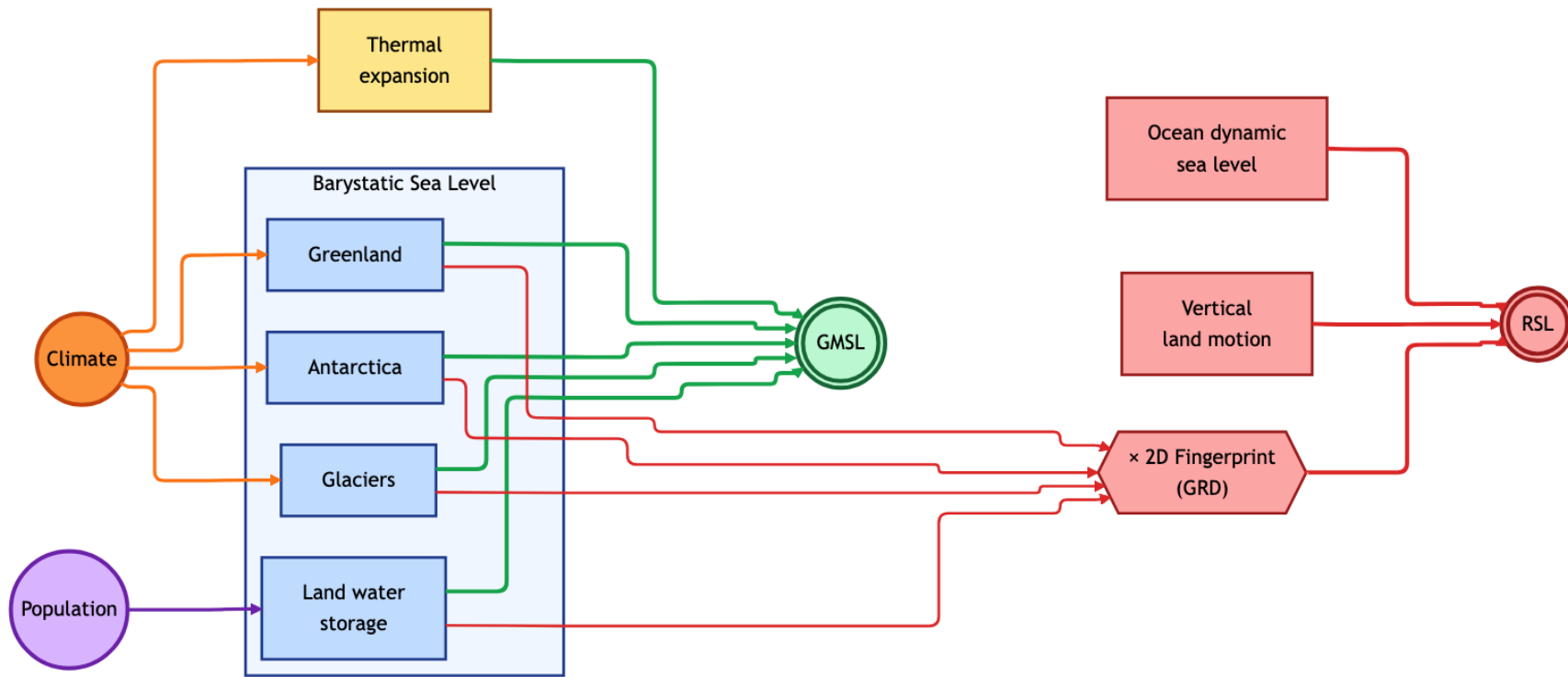
1. Take Historic observations from annual-mean tide-gauge records.
2. Apply a **Gaussian-process model** to those historical data points.

Isolates the non-climatic, long-term signals such glacial isostatic adjustment (GIA) from short-term ocean noise.

3. Estimate the historical background rate, that rate is **modeled linearly into the future**.

It is assumed that these long-term geological processes will continue at the same steady pace through 2100 and beyond.

Framework Design



“Workflows” in FACTS

Workflow	AIS	GrIS	Glaciers	
wf1e	emulandice	emulandice	emulandice	 Total 1
wf2e	LARMIP-2	emulandice	emulandice	 Total 2
wf3e	DeConto21	emulandice	emulandice	 Total 3
wf1f	AR5-like	FittedISMIP	AR5 Glaciers	 Total 4
wf2f	LARMIP-2	FittedISMIP	AR5 Glaciers	 Total 5
wf3f	DeConto21	FittedISMIP	AR5 Glaciers	 Total 6
wf4	Bamber19 SEJ	Bamber19 SEJ	Bamber19 SEJ	 Total 7

You can think of each workflow as a total sea-level from **summing over**
a specific combination of modules

medium / Low confidence Projections

Workflow	GrIS	AIS	Glaciers	Land water	Sterodynamic	VLM
Medium-confidence workflows						
1e	emulandice	emulandice	emulandice	ssp	t1m	kopp14
1f	FittedISMIP	ipccar5	ipccar5 (GMIP2)	ssp	t1m	kopp14
2e	emulandice	larmip	emulandice	ssp	t1m	kopp14
2f	FittedISMIP	larmip	ipccar5 (GMIP2)	ssp	t1m	kopp14

Low-confidence workflows						
3e	emulandice	deconto21	emulandice	ssp	t1m	kopp14
3f	FittedISMIP	deconto21	ipccar5 (GMIP2)	ssp	t1m	kopp14
4	bamber19	bamber19	ipccar5 (GMIP2)	ssp	t1m	kopp14

Thank You for coming to my Lecture!!!

Ask me many questions, I may not know the answers just yet, but
happy to **think things through together!**

Thank you **Chris** for Organizing this workshop,

Since the past 3 days, I have come to realize how little I
know about sea-level rise

&

There is sooo much more to do (YaY ??)

To all my mentors, thank you for you time and support. It means a lot!